

FireAtlas: NASA Space Apps 2026 project blueprint

Challenge: Harmonization of MODIS and VIIRS Hot Spots

Research status: 26 September 2026

Suggested one-line pitch: From two generations of satellite fire observations to a comparable burning calendar, a global fire context map, and an experimental crew awareness simulator.

1. Decision and challenge fit

The published challenge asks for **a web application that harmonizes MODIS and VIIRS active-fire records into a burning activity calendar**. Users should inspect historical patterns, unusual conditions, and critical periods for a selected area. This is the scored core. Build and demonstrate it before adding an operational-looking phone interface. Reference: [official NASA challenge](#).

Deliver three linked views:

- 1. Global Fire Atlas:** geographic exploration of detected burning, vegetation context, weather-based fire potential, and observation freshness.
- 2. Burning Activity Calendar:** the centerpiece. Choose an area and see daily/monthly burning activity, seasonal baselines, anomalies, sensor contributions, and a reproducible data-quality explanation.
- 3. Crew Scenario Lab:** a simulated firefighter and commander view that demonstrates exposure alerts and route analysis, subject to clear data-age and communications limits. Its prototype is **not an authorized field safety system**.

The project should never label every hotspot a wildfire or every green pixel safe. Agricultural burning and other thermal anomalies are among the observations in the challenge. Historical hotspot activity, current fire observations, pre-ignition fire weather, and an incident-level route estimate answer different questions and must appear as distinct layers.

2. Product specification

Priority	Screen / function	What it does	Acceptance demonstration
P0	Global map	Shows global aggregates at low zoom and individual NASA FIRMS observations at high zoom, with time, sensor and observation age	Select an AOI from globe; click an observation and see original record, confidence and timestamp
P0	Harmonized calendar	Daily cells, monthly seasonal bars, same-month historical baseline, source toggle, area and year selector	Compare MODIS-only, VIIRS-only, and deduplicated unique observation days without a false sensor-change trend
P0	Evidence drawer	Explains hotspot vs fire event vs burned area, no-data, data age, UTC / local date, and harmonization method	Download AOI data and reproduce chart from exported rows
P0	Forest context	Vegetation land-cover mask and greenness anomaly with acquisition date	Show an NDVI change as vegetation condition, not an immediate fire danger instruction
P1	Fire-weather layer	NASA fire-weather index and its valid time, separate from active detections	A historically active region can still show low current weather risk and vice versa
P1	Incident replay	Replays a historical incident using only data that were available at each replay timestamp	Prevent information from future overpasses leaking into past decisions
P1	Commander demo	Team roster, location age, connectivity state, acknowledged incident polygon, human-set rally/safety-zone candidates	A disconnected crew member is marked unknown rather than assumed safe
P1	Crew demo	Current GPS, accuracy, data age, alert and manual SOS, with simulated hazard and an approved route	Replay triggers local alert and commander notification; crew can acknowledge but alert retains history
P2	SAR change card	Before/after radar backscatter change and optical cross-check for a chosen historical burn	Document acquisition dates, radar geometry and classification uncertainty
P2	Forecast ensemble	Scenario spread under changing wind, uncertainty envelope, multiple route options	Forecast cannot be represented as a measured fire boundary

User journey

Explore world → select forest AOI → inspect 2000–present sensor-aware calendar → view what is unusual for this month → open an existing fire / historical replay → launch a preloaded crew exercise → simulate a changed fire polygon or wind forecast → inspect alert propagation and route revision. NASA's already existing [FIRMS](#) and [Fire Event Explorer](#) should be acknowledged in the submission: the added value is a transparent cross-sensor calendar plus the educational incident bridge, not a claim to have invented fire mapping.

3. Data and access catalogue

Requirement	Dataset / source	Resolution or cadence and proper use	Access and caveat
Main historical record	FIRMS MODIS Collection 6.1 (Terra/Aqua)	Approx. 1 km hotspot pixels; historical archive since 2000	Archive / standard science quality for long trends; NRT for recent display only
Main recent record	FIRMS VIIRS 375 m (S-NPP, NOAA-20/21)	Smaller fire sensitivity; S-NPP records from 2012	Sensor availability differs by year; verify current availability by FIRMS API before querying
Last several days and area searches	FIRMS Area API tutorial	Original points include acquisition date/time, confidence, FRP, sensor	Register free MAP_KEY; cache server-side; world query can exceed 100,000 observations/day
Historic validation of affected surface	MCD64A1 / VNP64A1	Burned area, not a live perimeter	Delayed and may miss small burns; useful for retrospective checks
Vegetation condition	MOD13Q1 NDVI/EVI ; optional NASA HLS	MODIS 250 m / 16-day vegetation composite; HLS 30 m for selected local cases	Greenness is not live fuel moisture; optical clouds/smoke mask pixels
Forest / fuel context	NASA MCD12Q1 land cover ; optional US LANDFIRE	Annual broad land cover; LANDFIRE 30 m US fuels for pilot	Land cover is not a fuel behavior model; LANDFIRE does not cover the globe
Fire weather	NASA GFWED with MERRA-2 history / GEOS forecast, optionally GEOS-FP	FWI incorporates temperature, humidity, wind, snow and precipitation; coarse regional guidance	Locally calibrate classes; 0.25° is far coarser than a fireline; experimental forecast age shown
Rain and soil context	NASA GPM IMERG , SMAP L4 soil moisture	Broad moisture context, not a high-resolution crew safety input	Distinguish near-real-time vs final runs; respect 9 km SMAP scale
Terrain	NASA SRTM / NASADEM	Approx. 30 m slope / aspect for regional spread modeling	No detailed ground-level trail condition information
Radar change	NASA OPERA RTC-S1 (processed Sentinel-1), plus NISAR data if covered	Change from pre/post SAR acquisitions; smoke/cloud resistant	Radar burn change ≠ current heat ≠ traversable path; geometries, moisture and forest structure matter
Imagery / visual QA	NASA GIBS / Worldview	Time-aware map tiles and comparison layers	Basemap imagery observation date must be shown
Event and firefront comparison	NASA-partner FEDS , NASA Fire Event Explorer	Estimated event/perimeter/front from VIIRS at roughly 12-hour cadence	Limited geography and latency; not an authorized operational perimeter
Paths and confirmed incident data	OpenStreetMap trails/roads, approved incident GIS polygons, commander field input	Non-NASA contextual inputs allowed if NASA drives the scientific question	Trails may be absent or blocked; obtain agency-provided perimeters for validation

Use an AOI query for API calls; obtain long records via the archive. Store original provenance and processing versions. Confirm exact collection versions, region coverage, key quotas, licenses and API response fields before implementation. NASA notes its recent NRT fire records can later be replaced with standard processed records; version and refresh the history accordingly.

4. Actual MODIS-VIIRS harmonization protocol

Raw data contract for each detection: `source_id`, `platform`, `sensor`, `product_version`, `processing_level`, `acquisition_utc`, `retrieval_time_utc`, `lon`, `lat`, `scan_m`, `track_m`, `confidence_raw`, `frp_raw`, `daynight`, `thermal_anomaly_flag`, `source_uri`. Preserve originals. Do not equate numeric MODIS confidence with categorical VIIRS confidence. Use explicit source-specific mappings only after study of both product guides. Keep FRP sensor-specific in the interface unless calibration is verified.

1. Select AOI and land-cover stratum; keep forest/shrub/grass/cropland views separate. Mask likely static industrial sources where possible; keep an “other/uncertain” group.
2. Convert acquisition times to UTC internally; convert to the AOI's local civil calendar only for display. International date line and regional time zones need deliberate handling.
3. Map observations to a common **analysis grid** (start at 1 km for local view and coarser equal-area cells for world view). Retain each original sensor pixel footprint and grid coverage. Never upscale a MODIS observation to claim VIIRS-like precision.
4. On a chosen day, group nearby detections in an observation-time window and intersect sensor footprints to identify candidate duplicate observations. For overpasses hours apart, keep repeated detections as separate observations but link them to the same candidate event. Set distance/time tolerances through an annotated overlap study rather than an arbitrary universal threshold.
5. Publish three series: **observed active-fire cell-days**, **candidate event-days**, and **sensor-specific raw pixel counts**. Label cell-days as a sampling proxy, not unique fire count or burned area. The default daily calendar can count each observed common-grid cell once even when multiple sensors detect it; report all contributing sensors.
6. During MODIS-VIIRS overlap (2012 onward), compare paired passes by biome, day/night, season and footprint; derive empirical sensor sensitivity bands. Keep a stable **MODIS-only historical reference** for 2000 onward and a **joint/VIIRS-era series** for 2012 onward. A fully normalized stitched time series is a research extension, not a prerequisite for a truthful MVP.
7. For science-grade exposure-adjusted rates, ingest companion fire-mask/cloud/coverage products or suitable Level-3/CMG summaries. With point-only FIRMS data, “no detections” does not mean “observed and no fire.” Display missing/uncertain observation coverage instead of dividing by an invented number of clear overpasses.
8. Derive an area-specific seasonal baseline only from comparable years and source configurations. Show anomaly as a percentile or $(\text{current} - \text{baseline_median})/\text{IQR}$, clearly labeling the baseline. Calculate percentiles from prior years only in an incident replay.

Human-check example: A forest cell with 3 MODIS pixels and 11 VIIRS pixels on one day is not 14 fires. A calendar count of one common-grid cell-day plus source provenance is defensible; a distinct fire-event count requires a separately validated event-tracking algorithm. Test against periods before/after VIIRS introduction so a sensor upgrade is not presented as a fire surge.

5. Risk model: separate scales and claims

Regional pre-ignition fire potential: NASA GFWED Fire Weather Index plus vegetation / land-cover context and a locally calibrated percentile. Weather conditions alone do not predict ignition. Global map: 25-km-ish contextual grids with original scale visible, not high-resolution false precision.

Incident observation: FIRMS points and optional FEDS estimated polygons; each has observation time, processing delay, uncertainty/quality, and the last successful fetch. Add credible agency perimeters when available and label the source. FIRMS pixels are not perimeter lines. Smoke, clouds, canopy, overpass intervals, and geolocation error can obscure or displace detections.

Experimental spread scenario: After P0 works, use [ELMFIRE](#) on one US case with verified fuels (LANDFIRE), terrain (SRTM), initial perimeter, weather forecast and wind perturbations. Output distribution of arrival times / envelope for 1–6 hours. A lower-complexity first demo may use scripted spread scenarios; label them simulated. AI may rank scenarios, detect anomalies and explain provenance, but must not invent a precise firefront or make sole evacuation decisions. Do not extrapolate a global 25 km weather grid to meter-scale hazard zones.

Crew zone map (exercise only): red = inside commander exclusion polygon or modeled immediate exposure; orange = close to a modeled/confirmed boundary or a route with insufficient time margin; yellow = monitored area; green = relatively lower modeled exposure; gray = missing, stale or contradictory inputs. Only a commander can approve actual incident boundaries, safe destinations and routes. Define numbers for spatial buffers / lead times with incident specialists; no universal distances are justified by satellite data.

Route analysis: Start from a trail / road graph or reviewed local grid. Remove confirmed closed edges and edges entering red/gray regions. Assign travel time accounting for slope and walking profile; compare conservative arrival time at each edge against the earliest fire arrival across spread scenarios plus a safety margin. Compute feasible time-dependent paths to **commander-designated** safe zones; rank by maximum time margin and exposure, then travel time. If no route survives, display **NO VALIDATED ROUTE — FOLLOW COMMAND / RADIO PROCEDURES**. SAR change layers can inform postfire terrain context after review; radar pixels cannot establish a passable footpath. The [NWCG LCES guidance](#) puts lookouts, communications, escape routes and safety zones first.

6. Phone and commander design

Deliverable at hackathon: an installable PWA for exploration plus an **Android native/Capacitor proof of concept** for local notification, foreground location, haptics, offline incident snapshot, and commander updates. A browser-only PWA cannot be assumed to deliver continuous background GPS and maximal loud alarms in every device state. iOS does not expose a general app-controlled system-volume setting; its mute-bypassing critical alerts require a special entitlement. Android location foreground services and background permissions are constrained. Design a standard high-priority audio + vibration alert, test on real devices, and show alarm volume status to the user; optional paired hardware buzzer later. Do not promise an uninterruptible forced-volume buzzer.

On device, calculate geofence intersections locally from **last approved incident snapshot** with GPS accuracy radius; send signed location/health pings to server when online. Commander receives a distinct alert with crew identifier, last location, age, accuracy and acknowledgment state; nearby teammates are informed only when online and incident policy authorizes it. Rate-limit repeated alerts and make red alert require affirmative acknowledgment (do not wait for user to leave red to stop every sound). If the device loses coverage, cache the incident and show **data unavailable; radio procedure**. Commander must see last-seen time, not a false live dot.

Minimum data protection: incident-scoped accounts, least-privilege roles, encrypted transport and at-rest crew location, no public crew feed, short configurable location retention, audit of commander

polygon changes and alert acknowledgments, local data wipe after exercise. Emergency messaging cannot depend on AI text generation; use deterministic templates with a radio confirmation fallback.

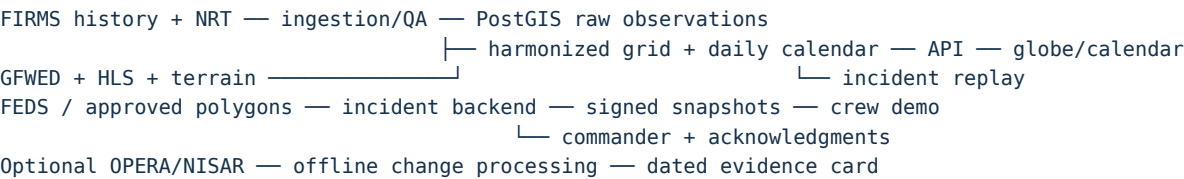
7. Repositories and research worth using

Resource	Direct benefit	Work / caution
NASA ARSET FIRMS notebook and FIRMS API tutorial	Working query/provenance example	Obtain MAP_KEY; cache, paginate as appropriate
NASA GIBS web examples	Time-aware NASA image layers	Overlay analytical results, do not scrape screenshots
Fireatlas/FEDS and NASA IMPACT Fire Event Explorer	Reference event tracking and reusable UI/data concepts	Its VIIRS event/perimeter processing does not deliver cross-sensor historical harmonization; check the respective licenses before copying code
NASA HLS tutorials	Greenness / burn-change case study	Select pre/post cloud-free scenes
OPERA application notebooks and ASF HyP3 SDK	Radar access, terrain correction and change example	Use postfire/landscape change; not live escape routing
ELMFIRE	Validated structure for future local spread scenarios	Complex inputs; reserve for after P0 and single region
MapLibre GL JS , deck.gl	Global map, time animation, fire pixels	Pre-aggregate world counts server-side
NetworkX / GRASS GIS r.cost	Experimental hazard-weighted trail and raster routes	Review connectivity and traversal; shortest geometric distance is inadequate

Research reading sequence:

1. Schroeder, Oliva, Giglio & Csiszar (2014), [VIIRS 375 m active-fire algorithm](#): smaller-fire response and fundamental sensor differences.
2. Giglio, Schroeder & Justice (2016), [MODIS Collection 6 active-fire algorithm](#): MODIS error sources and product meanings.
3. Field et al. (2015), [Global Fire Weather Database](#): weather-index basis and need for regional calibration.
4. Chen et al. (2022), [VIIRS event tracking / FEDS](#): benchmark event shapes and replay.
5. Giglio et al. (2018), [MODIS burned area algorithm](#), plus [VIIRS burned-area introduction](#): independent retrospective check.
6. Lautenberger (2013), [ELMFIRE original formulation](#): optional spread model.
7. [NASA NISAR fire-management example](#) and [OPERA notebooks](#): radar application and its time scale.

8. Architecture and incremental build



Use Python (FastAPI, Pandas/GeoPandas/Rasterio) for ETL and APIs; PostGIS for raw events, AOIs and spatial aggregates; a single common analysis grid (H3 or equal-area cells with explicit cell area); React + MapLibre + deck.gl for web; Android Kotlin or Capacitor plugin for the phone exercise. Use live NASA endpoints only on the server, not from each browser; keep API keys private. Prefetch historical aggregates as Parquet and tiles. Real-time commander updates can use WebSockets; disconnected clients use cached data and explicit timeouts.

Data contract examples

- `/api/calendar?bbox=...&year=...&stratum=forest&metric=cell_days`: date, count, baseline, anomaly, sensors, observation_coverage, provenance, source_version.
- `/api/hotspots?bbox=...&since=...`: raw records plus age; never convert to perimeter silently.
- `/api/incident/{id}/snapshot`: versioned approved polygons, candidate/approved routes, timestamp, expiry, signatures, field status.
- `/api/crew/{id}/ping`: role-gated encrypted location, accuracy, timestamp, connectivity; no public access.

Work sequence

Before hackathon (research and sample prep): obtain FIRMS key; choose a pilot AOI such as California's 2024 Park Fire (rich fuels and archival records); prepare a second contrasting agricultural-burning AOI so classification matters. Save dated subsets of 2005–2025 MODIS and 2012–2025 VIIRS; choose an overlapping few-month slice for calibration. Download a small dated NISAR/OPERA example if suitable coverage exists. Verify independent agency fire perimeters and redistribution terms.

First 8 hours: ingestion + clean schema + common analysis cells; generate two-sensor daily calendar; demonstrate 2000 vs 2012 baselines without a false jump. Freeze raw sample files and provenance.

Next 8 hours: implement AOI globe, timeline/calendar, zoom states, weather layer, vegetation layer, evidence drawer, export. Add smoke/cloud/no-data and agricultural/other uncertainty labels.

Next 6 hours: replay one historical incident with commander + simulated crews, scripted polygons, local alert and safe-route prototype. Then add optional FEDS or SAR evidence if time remains.

Final 2+ hours: three reproducible demos (seasonal calendar, current map, simulated crew danger); test airplane mode, stale GPS, stale satellite feed, missing trail, wind reversal, and two same-day sensors; record a short walkthrough and publish source/provenance/limitations. Prioritize a reliable calendar over unfinished forecasting.

9. Validation and success criteria

Claim	Check	Go / no-go interpretation
Harmonized record is honest	Review annotated same-day co-detections by biome, validate merged cell-days manually, compare 2011–2014 trend boundary	If sensor bias not constrained, show dual series instead of a false seamless trend
Calendar helps users	Ask a user to identify a local high-activity season and an anomalous month, then explain the supporting sensor data	They should distinguish historical activity from a prospective fire forecast
Wildfire-like classification	Compare sample forest/crop/industrial flags to land cover and local incident record	Do not claim exact global wildfire counts from all thermal detections
Incident replay is causal	At each replay date, hide later detections, final burned area, and later FEDS perimeters	No future data in an alert or routing decision
Model zones are well-calibrated	Compare hindcast envelope to held-out next perimeter; measure missed expansion and over-warning across multiple events	A single selected success story is insufficient for field claims
App fails visibly	Inject GPS error, location staleness, server outage, no signal, low battery, and closed trail	Show gray/unknown and radio/commander fallback; never silently show green
Alerts work in a demo	Test real Android audio/haptics and commander receipt while connected, then disconnected local geofence	Report observed latency on tested devices; no guarantee of alarm delivery

10. Risks, safeguards and differentiation

Scientific: VIIRS detects more small fire pixels than MODIS; new sensors and varying overpass coverage create false trends; active pixels have no cloud/non-observation denominator. Remedy: source-specific records, comparable overlap periods, QA masks and explicit missing status. **Forecasting:** FWI is a broad potential metric; landscape fire predictions depend on local fuels and wind. Remedy: scenario envelopes and locally validated thresholds. **SAR:** cloud/smoke penetration provides landscape change, not continuous live fire images. NISAR entered science operation in 2026 and provisional public products exist, but nominal repeat sampling is on the order of days. Remedy: show dated postfire change, keep it out of immediate route clearance. **Human safety:** a smartphone can fail through heat, smoke, no signal, flat battery, or imprecise GPS. Remedy: prototype exercise, human authority, offline last-approved snapshot, route expiry and LCES communication.

The winning claim is specific: **a transparent, comparable MODIS/VIIRS burning calendar that reveals where and when unusual activity occurs, then explains how that context can support a training scenario for fire crews.** Make the judge see the actual harmonization (source switch, duplicates, overlap bias and observation gaps). A globe alone exists already, and FEDS already provides VIIRS-derived event shapes; the explicit cross-sensor reasoning and trustworthy communication of uncertainty are the distinctive work.

11. Suggested demo and submission copy

1. Open the globe and click two contrasting places: forest and agricultural landscape. Show the actual NASA source/time and avoid claiming that all hotspots are wildfires.
2. Draw the pilot AOI and reveal 20+ years of MODIS history versus the 2012+ VIIRS era. Toggle sources. Show one season and one anomaly, explaining an observation gap and a deduplicated day.
3. Open a recent/historical incident. Show dated FEDS/agency evidence, GFWED weather context, and dated SAR burn change if available.
4. Tap **Start exercise**. Move a simulated crew member near a commander-defined exclusion area, trigger on-device + commander alerts, then invalidate the route by changing wind or closing a trail. Show **unknown** when the connection drops.
5. Finish with what the tool proved (calendar/replay) and what remains research (forecast calibration, device reliability, incident authorization).

Suggested title: FireAtlas: A Harmonized Burning Calendar and Crew Awareness Lab. **Tagline:** Know the season. See the evidence. Practice the response.

Source notes

Core limitations: [NASA FIRMS interpretation FAQ](#), [FIRMS active-fire meaning](#), [FEDS service metadata and limitations](#). Radar status: [NISAR mission data](#), [NISAR mission sampling](#). Alert platforms: [Android foreground location](#), [iOS critical alerts](#), [iOS volume control](#). Firefighter procedure: [NWCG LCES](#).